

Vending Machine--- Communication Protocol

For Cash or Card Payment Integration

1. the communication method

UART serial communication, parameter configuration

Baud rate	9600bps
Data bit	8 bits
Stop bit	1 bit
Check digit	none
Flow control	none

Data frame format :

ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGTH	DATA	CHK
1 Byte	1 Byte	1 Byte	1 Byte	1 Byte	n Bytes	1 Byte

[Description] :

1) [Address] is fixed to 0xFF

2) [frame number] 0x00;

3) [Data Header] When the Android APP is launched, it is 0x55, and when the vending machine device is launched, it is 0xAA;

4) [Data] low position first, followed by DATA0, DATA1...DATAn;

5) [Verification] The verification mode is accumulation and verification: that is, the entire data frame is accumulated and all other bytes except [address] [frame number] [CHK], and the result of lower 8-bit data is taken;

2. Command details :

1. Android APP to get device ID

	ADDR	FRAME NUMBER	HEAD ER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x31	0x01	0xAD	CHK
Device	0xFF	0x00	0xAA	0x31	0x0F	deviceID(15)	CHK

2. Android APP initiates control commands

2.1 Delivery Command

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x41	0x02	Lane(1) Delivery Quantity(1)	CHK
Device	0xFF	0x00	0xAA	0x41	0x02	Lane(1) Delivery Quantity(1)	CHK

【NOTICE】：Place an order and issue a delivery instruction, the device receives the instruction to deliver, and returns the data

2.2 Payment instructions

	ADDR	FRAME NUMBER	HEADE R	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x11	0x05	Pay amount(4) Payment method(1)	CHK
Devi ce	0xFF	0x00	0xAA	0x11	0x01	0x01	CHK

【NOTICE】：APP places an order and sends an CASH payment instruction, and the device receives the instruction and responds

Payment amount: 4 bytes of data, value range 0-4294967295, amount = commodity price × 100

Payment method: 1 byte of data, the value 0 represents the cancellation of the payment, the value 1 represents the coin or banknote waiting for payment, the value 2 represents the credit card waiting for payment

2.3 Android APP cancels the device failure instruction, and the device returns to the idle state

	ADDR	FRAME NUMBER	HEADER	CM D	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0xA 2	0x01	0xFF	CHK
Dev ice	0xFF	0x00	0xAA	0xA 2	0x01	0x01	CHK

【NOTICE】：APP remove the device failure, the APP successfully receives the message and returns 0x01

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2.4 Android APP issues a refund instruction

	ADD R	FRAME NUMBER	HEADE R	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0xB1	0x01	0xFF	CHK
Device	0xFF	0x00	0xAA	0xB1	0x01	0x01	CHK

【NOTICE】：The device returns 0x01 to indicate that the refund is successful

3.Android APP initiates status query

APP query device status frame

	AD DR	FRAME NUMBE R	HEA DER	CM D	DATA LENGHT	DATA	CHK
AP P	0xF F	0x00	0x55	0xE1	0x02	lane(1) Delviery quantity(1)	CHK

Device returns status data frame：

- Delivery completed

	AD DR	FRAME NUMBER	HEA DER	CM D	DATA LENGH T	DAT A	CHK
Devi ce	0xF F	0x00	0xAA	0xE1	0x01	0x01	CHK

- Equipment failure

	AD DR	FRAME NUMBER	HEA DER	CM D	DATA LENGH T	DAT A	CHK
Devi ce	0xF F	0x00	0xAA	0xE1	0x01	error code	CHK

error code：

0x02： lane motor failure

0x03： Device eye-light failure

- Payment completed

	AD DR	FRAME NUMBER	HEADER	CMD	DATA LENGH T	DATA	CHK
Devi ce	0xF F	0x00	0xAA	0xE1	0x04	Payment amount(4)	CHK

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【NOTICE】 : Payment amount: 4 bytes of data, value range 0-4294967295, amount = payment amount × 100

【NOTICE】 :When payment is made by credit card, the returned payment amount is equal to the commodity price×100

【NOTICE】 : When using coin/banknote payment offline, the returned payment amount is equal to the equipment balance

1) Android APP to check device balance

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0xE1	0x04	0x00(4)	CHK
Device	0xFF	0x00	0xAA	0xE1	0x04	balance(4)	CHK

【NOTICE】 : Balance: 4 bytes of data, value range 0-4294967295, amount=balance×100

2) Android APP to check the status of the coin change device

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x07	0x01	0x01	CHK
Device	0xFF	0x00	0xAA	0x07	0x01	status(1)	CHK

【NOTICE】 : The device returns to the current status of the coin refunder 0 or 1, 0 means that the coin can be returned, 1 means that the coin is insufficient and cannot be returned

【REMAK】 :

◆ **The complete processing flow for successful shipment is as follows :**

◆ APP query device balance instruction

0xFF	0x00	0x55	0xE1	0x04	0x00000000	CHK
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◆ Instructions to query the status of the coin change device

0xFF	0x00		0x55	0x07	0x01	0x01	CHK
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◆ APP receives the reply data frame of the coin change device

0xFF	0x00	0x55	0x07	0x01	status(1)	CHK
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◆ Return status 0 means that the coin can be refunded, and status 1 means that the coin is not enough for the coin to be returned.

◆ APP issue instructions to remove equipment failure

0xFF	0x00	0x55	0xA2	0x01	0xFF	CHK
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◆ APP received the instruction and returned success

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0xFF	0x00	0xAA	0xA2	0x01	0x01	CHK
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- ◆ APP send payment instruction data frame and wait for cash and cashless payment

0xFF	0x00	0x55	0x11	0x05	Pay amount(4) Payment method(1)	CH K
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- ◆ APP send query payment completion data frame

0xFF	0x00	0x55	0xE1	0x02	lane(1) Delivery quantity(1)	CH K
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- ◆ The device responds to the payment success data frame

0xFF	0x00	0xAA	0xE1	0x04	Pay amount(4)	CHK
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- APP send out shipping instruction data frame

0xFF	0x00	0x55	0x41	0x02	lane(1) Delivery quantity(1)	CH K
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- The device responds to the data frame after successfully receiving the APP delivery instruction

0xFF	0x00	0xAA	0x41	0x02	Lane (1) Delivery quantity(1)	CH K
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- APP query data frame of device delivery status

0xFF	0x00	0x55	0xE1	0x02	lane(1) Delivery quantity(1)	CH K
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- The device reports status information data frame

0xFF	0x00	0xAA	0xE1	0x01	Delivery completed/ Error code	CHK
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- APP issue a refund instruction

0xFF	0x00	0x55	0xB1	0x01	0xFF	CHK
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- The APP receives the device's response and the refund is successful

0xFF	0x00	0xAA	0xB1	0x01	0x01	CHK
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The entire delivery process of the data frame ends